

Vietnamese & English RAG Assistant

Project Overview

A full-stack Retrieval-Augmented Generation (RAG) application designed to process, index, and query multi-source documents in both Vietnamese and English. Built with modern async architecture to provide instant, grounded answers with exact source citations.

Core Features

- Multi-modal Ingestion: Supports seamless uploading of PDFs and crawling of website sitemaps.
- Intelligent Chunking: Advanced text processing and vectorization using Google Gemini Embeddings.
- Grounded AI Chat: Answers questions purely based on indexed documents with precise citation tracking.
- Real-time Sync: WebSocket-powered UI updates for document indexing status, system health, and chat latency.

Technical Stack

- Backend: Python, FastAPI, AsyncIO
- AI & ML: Google Gemini AI API (Embeddings & Generative Models)
- Database: Qdrant Vector DB, PostgreSQL (Relational Data), Redis (Pub/Sub & Job Queue)
- Infrastructure: Docker & Docker Compose
- Frontend: Vanilla HTML/CSS/JS with custom Glassmorphism design

Modern UI/UX Design

Features a custom-built, responsive "Glassmorphism" interface with:

- Native Dark/Light mode support
- Smooth micro-animations
- A highly optimized layout for analyzing sources alongside AI responses.

Why Choose This Gig?

- Production-Ready Code: Structured following best practices for scalability and maintainability.
- Bilingual Support: Optimized for both Vietnamese and English natural language processing.
- End-to-End Delivery: From infrastructure deployment (Docker) to frontend aesthetics.

This document serves as a portfolio showcase of my technical expertise in building advanced AI and Full-Stack applications.